Notes: Brandt, Van Biesebroeck, and Zhang (JDE, 2012)

Creative Accounting or Creative Destruction? Firm-level Productivity Growth in Chinese Manufacturing

Contents

1 Big picture

- **Question.** How large was productivity growth in Chinese manufacturing between 1998 and 2007, and how much of that growth came from incumbents, entry and exit, and reallocation?
- **Setting.** The paper studies an unbalanced panel of Chinese manufacturing firms drawn from the National Bureau of Statistics' annual survey of above-scale industrial firms. The period spans China's WTO accession and intense restructuring of the state sector.
- **Core challenge.** Measuring productivity in China is difficult because the raw data require extensive cleaning: deflators must be built, firm IDs change after restructuring, labor compensation appears understated, and capital stocks are reported at original purchase prices rather than comparable real values.
- **Main empirical result.** For the authors' preferred estimates, weighted average annual productivity growth for incumbents is 2.85% under a gross-output production function and 7.96% under a value-added production function over 1998–2007.
- **Main aggregate result.** Productivity growth at the manufacturing-sector level is even larger because net entry contributes strongly. Over the full sample period, net entry accounts for more than two-thirds of total TFP growth.
- **Main decomposition result.** TFP growth dominates input accumulation as a source of manufacturing growth and explains roughly two-thirds of labor-productivity growth.
- **Important qualification.** China exhibits massive gains through entry and exit, but much weaker productivity-enhancing reallocation among continuing firms than the United States. So the extensive margin is dynamic, while the intensive reallocation margin remains constrained.
- **Why this paper matters.** It is a benchmark study of Chinese manufacturing productivity using micro data. It links firm-level measurement, creative destruction, ownership change, and aggregate growth accounting in one coherent framework.

2 Data and measurement (key objects)

2.1 Observed firm-level panel

- **Data source.** Annual surveys conducted by China's National Bureau of Statistics for 1998–2007.
- **Coverage rule.** The survey includes all state-owned industrial firms and all non-state industrial firms with annual sales above 5 million RMB. Industry here includes mining, manufacturing, and public utilities; the analysis in the paper restricts attention to manufacturing.
- **Economic coverage.** The panel represents about 90% of manufacturing gross output. Although around 80% of all industrial firms are excluded by the above-scale cutoff, those omitted firms account for only a small fraction of economic activity.
- **Manufacturing sample size.** The unbalanced manufacturing panel rises from 148,685 firms in 1998 to 313,048 in 2007.
- **Dynamics of the panel.** The average annual attrition rate is slightly below 14%. Entry averages nearly 20% per year, more than offsetting exit.

- **Survival.** Out of the 148,685 firms in the 1998 manufacturing sample, only 33,054 survive through 2007.
- **Summary scale.** For the full industrial sample in Table 1, value added rises from 1.94 to 11.70 trillion RMB between 1998 and 2007, exports from 1.08 to 7.34 trillion RMB, and employment from 56.44 to 78.75 million workers.

2.2 Matching firms over time

- **Problem.** Firms sometimes receive new official IDs because of restructuring, merger, or acquisition. If these changes are ignored, continuing firms can be misclassified as exits and entries.
- **Solution.** The authors build longitudinal links using firm IDs when possible and otherwise use names, industry, addresses, and related identifiers.
- **Match rates.** Overall, 95.9% of year-to-year matches use official IDs and 4.1% use other identifying information.
- **Importance.** One-sixth of firms observed for more than one year experience at least one official ID change during the sample.

2.3 Observed variables and constructed variables

- **Core variables.** Gross output, value added, labor, wages and compensation, exports, and capital.
- **Labor compensation.** Employee compensation includes wages, supplementary benefits, unemployment insurance, retirement benefits, health insurance, and housing benefits.
- **Labor-share issue.** Even with these adjustments, labor's share of value added in the sample is only 34.2%, well below the roughly 55% seen in national accounts. In some exercises the authors inflate wage payments by a common factor to bring labor's share closer to the aggregate benchmark.
- **Capital-stock issue.** Firms report fixed assets at original purchase prices rather than as comparable real capital stocks. The authors therefore construct a real capital series using an adjustment procedure plus perpetual inventory.
- **Implication of capital adjustment.** Their preferred real-capital series implies capital per worker rose by 35.9% between 1998 and 2007, almost twice the growth rate implied by the raw official stock series.
- **Investment for proxy estimation.** Investment is not directly reported and is inferred from the capital-stock law of motion.
- **Sample cleaning.** Firms with missing or implausible values for key variables are dropped, as are firms with fewer than 8 employees because those firms operated under a different legal regime.

2.4 Unobservables and timing

- **Firm productivity.** Productivity is not directly observed and must be constructed relative to a production technology.
- **Entry definition.** Entrants in the event-study exercises are firms whose reported birth year is at most two years before they first appear in the sample.
- **Balanced versus unbalanced samples.** The paper repeatedly compares the full unbalanced sample with a balanced panel of firms active throughout the sample in order to isolate the contribution of entry and exit.

2.5 Empirical applications in the paper

The paper uses the same underlying panel to answer five related questions:

- How fast do continuing firms improve productivity?
- Where do entrants and exiters sit in the productivity distribution, and how do entrants evolve after entry?
- Do resources move toward more productive continuing firms?
- How much of aggregate manufacturing growth comes from TFP versus capital and labor accumulation?
- How do these patterns differ by ownership type and by sector?

3 Models and empirical specifications (all equations + interpretation)

3.1 Production function and the meaning of productivity

The paper starts from a generic production technology:

$$Q_{it} = A_{it} F_{it}(X_{it}). \tag{1}$$

Relative productivity can then be written as

$$\ln\left(\frac{A_{it}}{A_{j\tau}}\right) = \ln\left(\frac{Q_{it}}{Q_{j\tau}}\right) - \ln\left(\frac{F_k(X_{it})}{F_k(X_{j\tau})}\right). \tag{2}$$

- **Interpretation.** Productivity is always relative to a maintained technology k. To compare firms or compare a firm across time, one must evaluate both observations using the same input aggregator.
- **Why this matters.** The whole measurement exercise hinges on how aggressively one imposes common technology across firms and time.

3.2 Benchmark nonparametric measure: Törnqvist index numbers

The benchmark productivity-growth measure for a value-added production function is

$$\mathrm{TFPG}_{it}^{\mathrm{IN}} = (q_{it} - q_{it-1}) - \bar{S}_{it}(l_{it} - l_{it-1}) - \big(1 - \bar{S}_{it}\big)(k_{it} - k_{it-1}), \tag{3}$$

where $\bar{S}_{it} = (S_{it} + S_{it-1})/2$ is the average wage-bill share in value added.

To compare levels across firms in the same industry, the paper uses the multilateral productivity index

$$\ln \mathrm{TFP}_{it}^{\mathrm{IN}} = (q_{it} - \bar{q}_t) - \tilde{S}_{it}(l_{it} - \bar{l}_t) - \big(1 - \tilde{S}_{it}\big)(k_{it} - \bar{k}_t), \tag{4}$$

with $\tilde{S}_{it} = (S_{it} + \bar{S}_t)/2$.

- **Interpretation.** The Törnqvist approach does not require estimating technology parameters. Instead, local factor shares proxy for local substitution elasticities.
- **Main advantage.** It allows technology heterogeneity across observations and avoids placing strong functional-form restrictions on production.
- **Gross-output version.** For gross output, intermediate inputs enter exactly like another variable input, so the growth formula subtracts their growth weighted by the corresponding revenue share.

3.3 Parametric productivity measures: Olley-Pakes and Ackerberg-Caves-Frazer

For robustness, the paper also estimates sector-specific Cobb-Douglas production functions. In value-added form, a representative specification is

$$q_{it} = \hat{\alpha}_L^s l_{it} + \hat{\alpha}_K^s k_{it} + a_{it}^P, \tag{5}$$

so that productivity is the residual

$$a_{it}^P = q_{it} - \hat{\alpha}_L^s l_{it} - \hat{\alpha}_K^s k_{it}.$$

In gross-output form, an intermediate-input term is added as well.

- **Olley-Pakes (OP).** Uses investment as a proxy for unobserved productivity and models exit selection.
- **Ackerberg-Caves-Frazer (ACF).** Uses intermediate inputs as the proxy variable and identifies all coefficients in a second-stage GMM procedure.
- **Why the paper includes them.** These are standard structural estimators in the productivity literature and provide a check on the index-number results.

3.4 Why the benchmark remains the index-number measure

The parametric estimators produce higher productivity growth than the benchmark. The paper emphasizes two reasons:

- the estimated returns to scale are strongly decreasing, which mechanically boosts measured TFP growth in a rapidly growing economy;

- measurement error or markups may bias value-based production-function estimates downward in their input elasticities.

The average sum of labor and capital elasticities is about 0.80 under OP and barely 0.70 under ACF. The authors view those estimates as implausibly low and therefore focus on the index-number results as their benchmark.

3.5 Gross output versus value added

A central distinction in the paper is between value-added and gross-output production functions.

- **Value-added TFP** treats intermediate inputs as netted out from output.
- **Gross-output TFP** treats intermediates as direct inputs in production.

Because intermediate inputs absorb a large share of gross output, measured TFP growth under gross output is much smaller. In the sample, value added is only about 27–29% of gross output, and the benchmark gross-output TFP growth is 2.89% annually before the preferred downward adjustments.

3.6 Event-time regressions for entrants and exiters

To study how productivity evolves around entry and exit, the paper runs event-time regressions of productivity levels and growth rates on entry-year and post-entry dummies, as well as on exit-year and pre-exit dummies, including province and sector-year fixed effects. In reduced form, the idea is

$$\text{TFP}_{it} = \sum_h \beta_h \mathbf{1}\{\text{event time} = h\} + \lambda_{st} + \mu_p + u_{it}.$$

- **Interpretation.** The coefficients trace out where entrants start in the productivity distribution, how quickly they learn, and how far productivity falls before exit.

3.7 Reallocation among continuing firms

Following Bartelsman and Dhrymes, the paper compares the evolution of:

- an unweighted average productivity measure across continuing firms, and
- a weighted aggregate productivity measure using input or output shares.

- **Interpretation.** If resources systematically move toward more productive continuing firms, the weighted series should grow noticeably faster than the unweighted series.
- **Key empirical message.** This gap is large in the United States but tiny in China.

3.8 Aggregate growth decomposition

The macro-style growth-accounting identity is

$$\Delta \ln VA = \text{capital contribution} + \text{labor contribution} + \Delta \ln \text{TFP}.$$

To decompose aggregate productivity growth itself, the paper uses the Baily-Hulten-Campbell approach:

$$\ln \mathrm{TFP}_t = \sum_i \theta_{it}^Q \ln \mathrm{TFP}_{it},$$

where θ_{it}^Q is the gross-output share of firm i.

- **Interpretation.** Aggregate TFP changes can then be split into contributions from continuing firms and from net entry.
- **Alternative decomposition.** The paper also discusses the Petrin-Levinsohn decomposition, which removes the welfare effect of share changes among continuing firms when factor allocation is efficient.

3.9 Ownership and sectoral decompositions

Finally, the paper decomposes growth by ownership group and by sector.

- Ownership groups include state, private, foreign, privatized, and nationalized firms.
- Sectoral heterogeneity is summarized using a Harberger "sunrise" diagram, which plots cumulative productivity contributions against cumulative value-added shares across 2-digit industries.

4 Identification and measurement credibility (what makes the estimates believable)

4.1 What is *not* the source of identification

This paper is not a causal-policy paper built on an instrument or natural experiment. Its contribution is instead measurement and decomposition. The central question is whether the data, productivity estimators, and firm dynamics are treated carefully enough that the measured patterns are economically meaningful.

4.2 Why the benchmark approach is credible

- **Index-number benchmark.** The main estimates rely on Törnqvist index numbers, which require fewer functional-form assumptions than parametric production-function estimators.
- **Technology heterogeneity.** Because factor shares vary across observations, the index-number measure is less likely to confound productivity with an incorrectly imposed common technology.
- **Cross-method robustness.** OP and ACF yield similar qualitative patterns even when the levels differ.

4.3 Why the data work matters

- **Coverage validation.** The sample aggregates closely to the Chinese Statistical Yearbook and the 2004 Census for above-scale firms.

- **Firm matching.** Reconstructing links when IDs change is essential, because otherwise restructuring would look like spurious exit and entry.
- **Capital measurement.** Converting original purchase-price fixed assets into real capital stocks avoids systematic age and inflation bias.
- **Deflators.** The paper constructs firm-appropriate deflators for output, materials, and investment rather than relying naively on raw nominal values.

4.4 Why the robustness exercises are informative

The paper explicitly checks the sensitivity of TFP growth to four concerns highlighted in the China growth-accounting debate:

- undermeasured price inflation,
- understatement of labor input quality,
- incorrect factor shares,
- and mismeasured initial capital stocks.

The preferred estimates are deliberately conservative, combining three downward adjustments relative to the benchmark.

4.5 Relation to the literature

- Relative to **Young (2003)**, the paper argues that Chinese non-agricultural productivity growth has been understated, especially once manufacturing is isolated and firm-level data are used.
- Relative to **Hsieh and Klenow (2009)**, the paper agrees that potential gains from reallocation are large, but shows that actual productivity-enhancing reallocation among continuing firms is modest.
- Relative to **Olley and Pakes** and **Ackerberg-Caves-Frazer**, the paper treats proxy-based production-function estimation as useful robustness rather than as the benchmark.
- Relative to **Baily et al. (1992)** and **Foster, Haltiwanger, and Krizan**, the paper emphasizes the outsized role of net entry in Chinese manufacturing productivity growth.

5 Tables and figures

5.1 Table 1: Summary statistics on the underlying firm-level data set

Read:

- The sample is economically massive. For all industrial firms, value added rises from 1.94 to 11.70 trillion RMB between 1998 and 2007, while exports rise from 1.08 to 7.34 trillion RMB.
- Employment rises from 56.44 to 78.75 million, but the number of firms rises much faster, consistent with a wave of entry and restructuring.

- The table establishes that the paper is not studying a niche subsample. It covers the dominant part of Chinese industrial activity.

Table 1
Summary statistics on the underlying firm-level data set.

Year	Number of firms	Value added	Sales	Output	Employment	Export	Net value of fixed assets
1998	165,118	1.94	6.41	6.77	56.44	1.08	4.41
1999	162,033	2.16	6.99	7.27	58.05	1.16	4.73
2000	162,883	2.54	8.42	8.57	53.68	1.46	5.18
2001	169,030	2.79	9.24	9.41	52.97	1.61	5.45
2002	181,557	3.30	10.95	11.08	55.21	2.01	5.95
2003	196,222	4.20	14.32	14.23	57.49	2.69	6.61
2004	279,092	6.62	20.43	20.16	66.27	4.05	7.97
2005	271,835	7.22	24.69	25.16	68.96	4.77	8.95
2006	301,961	9.11	31.36	31.66	73.58	6.05	10.58
2007	336,768	11.70	39.97	40.52	78.75	7.34	12.34

Notes: all values are denoted in trillion RMB and employment in millions of workers. All industrial firms are included while the analysis in the paper is limited to firms in the manufacturing sector. A comparison with corresponding values in the China Statistical Yearbook, the China Statistical Abstract, and the 1995 and 2004 Census is in the online Appendix.

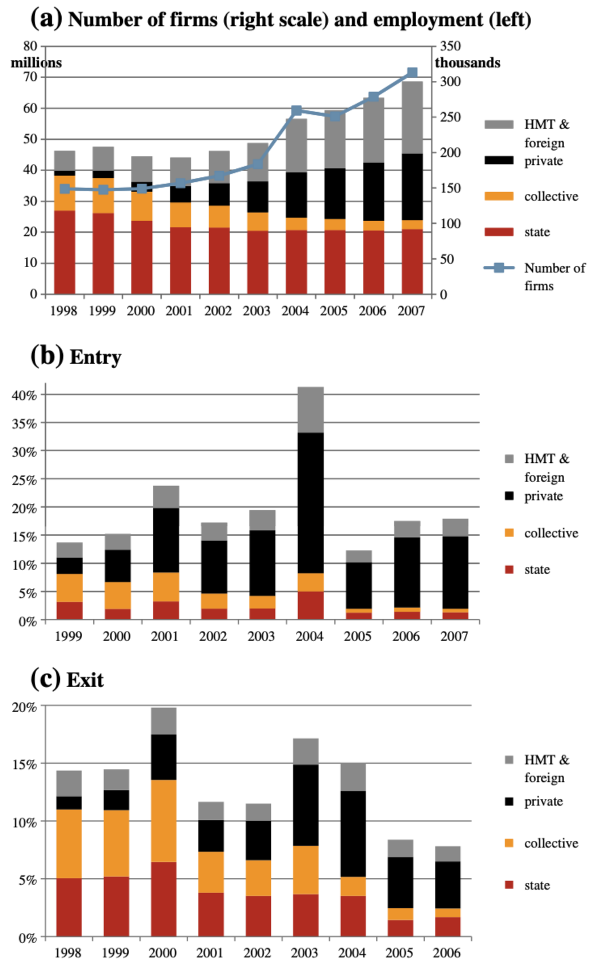


Fig. 1. Employment, entry and exit by ownership type.

5.2 Figure 1: Employment, entry, and exit by ownership type

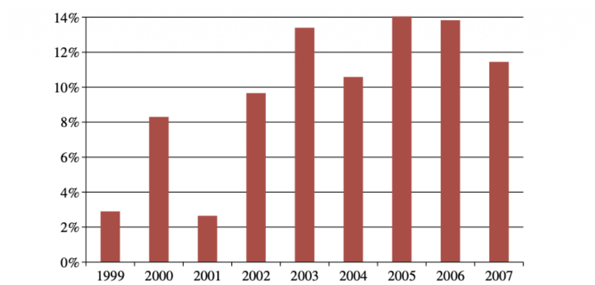
Read:

- The manufacturing landscape changes dramatically over time: private and foreign-related firms expand while state and collective ownership shrink in relative importance.
- Entry averages nearly 20% per year and exceeds exit, while attrition is still substantial.
- The visible jump in the number of firms between 2003 and 2004 reflects the industrial census and improved identification of firms that should have been in the sample earlier.
- This figure visually motivates why a decomposition through firm dynamics is central for China.

5.3 Figure 2: Benchmark firm-level TFP growth estimates

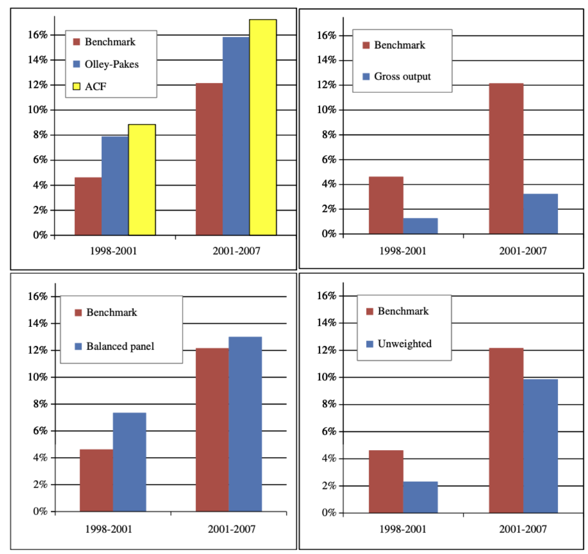
Read:

- Using the index-number method, value-added-weighted firm-level TFP growth rises from about 2.9% in 1999 to around 14% in 2005 before easing to 11.5% in 2007.
- Productivity growth is therefore both high and increasing over most of the sample.
- The timing lines up with post-crisis recovery and the WTO period, suggesting especially strong dynamism after 2001.



Note: Value-added weighted average of firm-level year-on-year TFP growth estimates obtained using the index number methodology on the full unbalanced sample of Chinese manufacturin firms.

Fig. 2. Benchmark firm-level TFP growth estimates.



Note: The benchmark estimates always use the index numbers to calculate productivity (instead of the Olley-Pakes parametric methodology); use value added weights (instead of unweighted averages); are calculated on the full sample of firms (instead of the balanced panel); and use a value added production function (instead of gross output). The alternative calculations in the four panels use the alternative assumptions indicated in brackets here.

Fig. 3. Alternative TFP growth estimates.

5.4 Figure 3 and Figure 4: Alternative TFP estimates and robustness

Read:

- OP and ACF estimates are above the benchmark, while gross-output productivity growth is much lower than value-added productivity growth.
- Unweighted firm averages are lower than value-added-weighted averages, implying larger Chinese firms are improving productivity faster than the average firm.
- The benchmark estimate is reduced by several robustness adjustments. In Figure 4, the 14-sector deflator yields 8.93%, using wage-bill growth as labor input yields 6.00%, unadjusted wage shares yield 8.77%, and the original-capital series yields 9.22%.
- The authors' preferred compromise estimate is 7.96% for value added and 2.85% for gross output.

5.5 Table 2: Average TFP growth in different countries

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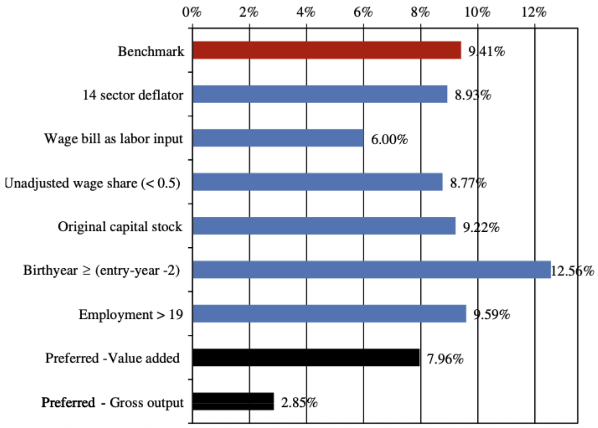


Fig. 4. Robustness of TFP growth estimates to a variety of assumptions.

Table 2
Average TFP growth in different countries.

Country	Study	Period	Sector	TFP growth
Firm-level: value added production function				
China		1998–2007	Manufacturing	0.080
Slovenia	De Loecker and Konings (2006)	1994–2000	Manufacturing	0.085
Vietnam	World Bank (2007)	2001–2003	Manufacturing	0.073
U.S.	Baily et al. (1992)	1982–1987	Selected manufacturing sectors	0.031
Chile	Pavcnik (2002)	1979–1986	Manufacturing	0.028
Firm-level: gross output production function				
China		1998–2007	Manufacturing	0.028
Korea	Ahn et al. (2004)	1990–1998	Manufacturing	0.035
Taiwan	Aw et al. (2001)	1986–1991	9 manufacturing sectors	0.021
Mexico	Tybout and Westbrook (1995)	1984–1990	Manufacturing	0.019
U.S.	Haltiwanger (1997)	1982–1987	Manufacturing	0.017
Japan	Ahn et al. (2004)	1994–2001	Manufacturing	0.003

Note: where published productivity growth estimates for other countries are available for several time periods, we have taken those from cyclical expansions to be comparable with the Chinese macroeconomic situation. The Chinese averages are the "preferred" estimates incorporating three adjustments on the benchmark estimates, as at the bottom of Fig. 4.

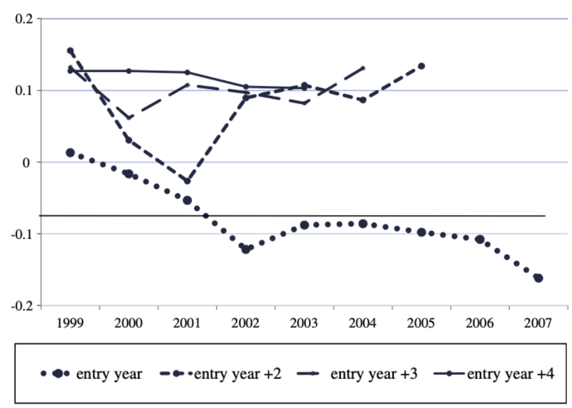
- China's preferred value-added TFP growth of 8.0% is comparable to Slovenia (8.5%) and above the United States (3.1%) and Chile (2.8%).
- In gross-output terms, China's 2.8% is comparable to Korea's 3.5%, above Taiwan's 2.1%, Mexico's 1.9%, and the United States' 1.7%, and far above Japan's 0.3%.
- The message is that China's performance is exceptional but still plausible when compared to high-growth benchmark episodes elsewhere.

5.6 Table 3 and Figure 5: Productivity evolution of entering and exiting firms

Table 3
Productivity evolution.

(a) Entering firms				(b) Exiting firms			
Relative to	Full sample		Balanced sample		Full sample		Balanced sample
	TFP growth (1)	TFP level (2)	TFP level (3)		TFP growth (4)	TFP level (5)	TFP level (6)
t_0 (entry)		0.031	−0.047	t_0 (exit)	−0.145	−0.141	−0.258
		(0.004)	(0.005)		(0.002)	(0.003)	(0.004)
$t+1$	0.139	0.147	0.073	$t-1$	−0.005	−0.079	−0.169
	(0.003)	(0.004)	(0.005)		(0.003)	(0.003)	(0.004)
$t+2$	0.016	0.130	0.058	$t-2$	−0.042	−0.065	−0.154
	(0.003)	(0.004)	(0.005)		(0.003)	(0.003)	(0.004)
$t+3$	−0.014	0.086	0.017	$t-3$	0.009	−0.019	−0.108
	(0.003)	(0.004)	(0.006)		(0.003)	(0.003)	(0.004)
$t+4$	−0.001	0.053	−0.022	$t-4$	0.024	−0.007	−0.091
	(0.004)	(0.005)	(0.006)		(0.004)	(0.004)	(0.004)

Note: OLS regressions of TFP level and growth rates on dummies for the entry and post-entry years, in columns (1)–(3), and exit and pre-exit years in columns (4)–(6). Province and sector-year fixed effects are included. These results use the index number estimates of productivity; comparable results using the Olley–Pakes and Ackerberg–Caves–Frazer productivity measures are in the Appendix.



Note: The markers represent coefficient estimates for productivity levels in post-entry years relative to balanced panel firms, comparable to those in column (3) of Table 3. Estimates are performed separately by entry cohort and the two-year moving average of coefficients is plotted. Productivity is estimated using the index number method; only recently-founded firms (at most 2 years ago) are counted as entrants; the year of entry and the first complete post-entry years are lumped together.

Fig. 5. Evolution of relative productivity levels for new entrants.

Read:

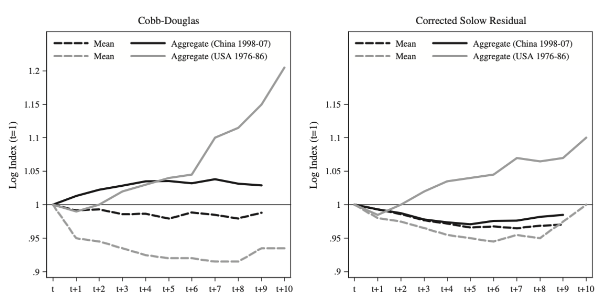
- Entrants do not enter with dramatically superior productivity levels. In the full sample they begin around the incumbent mean, but one year after entry they display a strong productivity advantage.

- In Table 3, the post-entry productivity-growth coefficient is 0.139 in year $t+1$, and the level coefficients remain positive for several years.
- Exiters are very weak performers near exit. In the final year of operation the TFP-growth coefficient is -0.145 and the TFP-level coefficient is -0.141 relative to the full sample; the balanced-sample gap is even larger.
- Figure 5 shows an important cohort fact: the initial productivity of entrants falls over time, but surviving entrants converge quickly to a stable premium over incumbents, usually by the third or fourth year.

5.7 Figure 6: Productivity contribution of resource reallocation between active firms

Read:

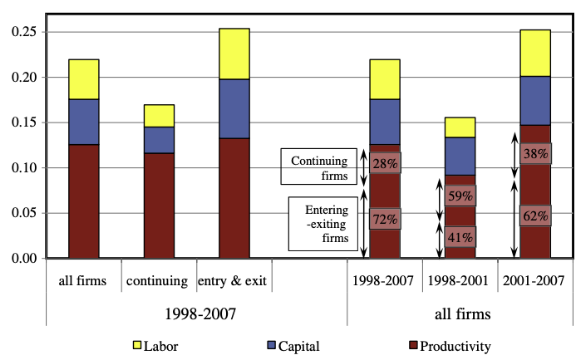
- In the United States, weighted aggregate productivity rises much faster than the unweighted mean, indicating strong reallocation toward more productive continuing plants.
- In China, that gap is tiny. With Cobb-Douglas residuals the cumulative gain from reallocation among active firms over nine years is only about 4%, and with the corrected Solow residual it is essentially zero.
- This is one of the paper's most striking facts: China is highly dynamic through entry and exit, but not through within-incumbent input reallocation.



Source: U.S. results are from Figure 1 in Bartelsman and Dhrymes (1998); for consistency, we apply the same methodology for Chinese firms.

Notes: Cobb-Douglas productivity measures are calculated from a production function estimated by least squares. Solow Residuals refer to a Tornqvist index number using the wage share as weight on labor input and the material share for materials; in the United States a "capital share" is observed, while in China constant returns to scale are assumed. Both productivity measures are purged from year and 4-digit industry effects, and residuals are used. The dashed lines are unweighted averages, the solid lines weigh plants or firms by their aggregate input ($L^{\alpha_L}M^{\alpha_M}K^{1-\alpha_L-\alpha_M}$) share.

Fig. 6. Productivity contribution of resource reallocation between active firms.



Note: The bars indicate the different components in a standard Solow growth decomposition showing average annual changes in log points. The weight on labor input growth is 0.42, the average of the firm-level sample average and that in the national accounts. The percentages on the right decompose productivity growth into changes at intensive and extensive margins, as discussed in the text.

Fig. 7. Output and productivity growth decompositions.

5.8 Figure 7: Output and productivity growth decompositions

Read:

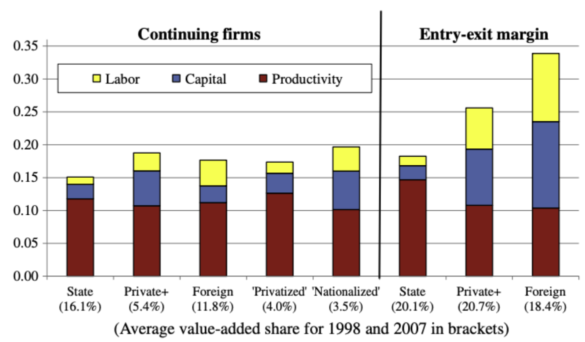
- Manufacturing value added grows at more than 22% per year over 1998–2007.
- Capital accumulation contributes about 5.1% per year and labor input about 4.5%; the remaining 13.4%, or 57% of output growth, is attributed to TFP.

- Net entry explains 72% of total productivity growth over the full sample, even though its output share is only 59%.
- Even in the short pre-WTO period, net entry still explains 41% of productivity growth, far above comparable U.S. figures.

5.9 Figure 8: Growth decomposition by ownership categories

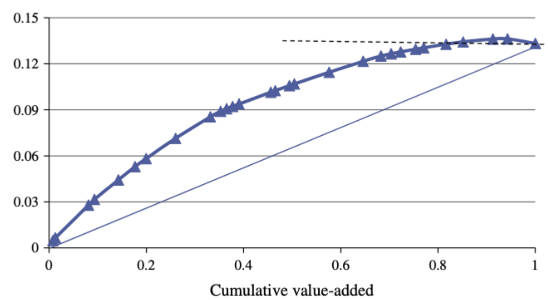
Read:

- Continuing state-owned firms actually post slightly higher TFP growth (12.5%) than continuing private (11.3%) or foreign (11.8%) firms, and privatized firms do even better at 13.5%.
- But output growth differs much more than TFP growth because the key difference is not just efficiency, but who receives labor and capital.
- At the entry-exit margin, newly appearing state firms produce almost five times as much value added as disappearing state firms even with only modest capital expansion and lower employment.
- The broad lesson is that ownership matters mainly through the allocation of new resources and the extensive margin, not through strong reallocation among continuing firms.



Note: Private+ combines collective-owned firms (and other hybrids) and purely private firms. The nationalized firms are those switching to majority state ownership, the privatized category are those switching from state to one of the other two or switches within the private cateogory.

Fig. 8. Growth decomposition by ownership categories (1998–2007).



Note: 2-digit sectorsare ordered from high to low productivity growth from left to right. The line plots the cumulative contributions to manufacturing productivity growth against cumulative value added.

Fig. 9. Harberger sunrise diagram for cumulative productivity growth.

5.10 Figure 9: Harberger sunrise diagram for cumulative productivity growth

Read:

- Chinese manufacturing productivity growth is broad-based rather than concentrated in just a handful of sectors.
- Sectors accounting for at least 85% of value added are needed to generate total aggregate TFP growth, far broader than the usual U.S. "mushroom" pattern emphasized by Harberger.
- Only one sector, tobacco processing, makes a negative contribution over the full period.
- Transport equipment and ordinary machinery are especially important contributors, while electric equipment and food processing are large sectors with relatively weak productivity growth.

6 Short summary

- Chinese manufacturing experienced extremely rapid productivity growth between 1998 and 2007, even under conservative measurement choices.
- The most important source of aggregate productivity growth was creative destruction: new firms entered at scale, exiters were relatively unproductive, and net entry contributed more than two-thirds of total TFP growth.
- The weak link is reallocation among continuing firms. China generated large extensive-margin gains, but much smaller within-incumbent resource shifts toward the most productive firms than the United States.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that Chinese manufacturing productivity growth in 1998–2007 was extremely high by international standards.
- It also establishes that aggregate manufacturing dynamism came disproportionately from entry and exit rather than from reallocation among surviving firms.
- Finally, it shows that a careful productivity analysis for China requires substantial data construction: matching firms over time, building real capital stocks, and stress-testing estimates against alternative measurement choices.

7.2 Economic intuition

Chinese manufacturing during this period looks like a system in which liberalization and competition made it easier for new firms to enter and for weak firms to disappear, but in which remaining distortions still limited the movement of capital and labor toward the most productive surviving producers. That combination generates a distinctive pattern: enormous gains from creative destruction, strong post-entry learning, and broad sectoral productivity growth, but relatively weak Olley-Pakes-style covariance gains within continuing firms.

7.3 Takeaway

This paper's most important lesson is that China's manufacturing miracle was not just a story of incumbent firms becoming more efficient. It was also, and perhaps even more, a story of who entered, who exited, and who was allowed to expand. If one wants to understand Chinese productivity growth, one must study firm dynamics and ownership transitions in addition to conventional growth accounting. At the same time, the paper makes clear that the next stage of reform lies not only in keeping entry and exit open, but in improving the within-industry allocation of resources among firms that remain active.